



A Reference Architecture for Digital Product Passports at Batch Level to Support Manufacturing Supply Chains

Malina Wiesner¹, João Moreira¹, Renata Guizzardi¹(✉), and Paul Scholz²

¹ University of Twente, Enschede, The Netherlands

{j.luizrebelomoreira,r.guizzardi}@utwente.nl

² Hilti Group, Thüringen, Austria

paul.scholz@hilti.com

Abstract. Despite the availability of metrics and measurement tools, the lack of formal models and standardization concerning product lifecycle information poses a challenge in assessing how sustainable the supply chain operations of a company are. Digital Product Passports (DPPs) emerge as a promising solution to track and ensure accurate product information is maintained throughout the whole product lifecycle. DPPs are digital representations of the accumulated information of a particular product from its inception to end-of-life. We investigated the topic from the perspective of a large European manufacturer, examining how different phases of the product lifecycle can be supported by static and dynamic data at different levels of granularity. As a result of our research, we propose a reference architecture that supports the development of DPPs with emphasis on product components at batch level granularity. The validation of the proposed architecture shows that the approach provides an opportunity for manufacturers to address sustainability issues in resource-intensive manufacturing supply chains while actively reusing legacy infrastructure.

Keywords: Digital Product Passport · Sustainable Manufacturing Supply Chain · Reference architecture · Interoperability

1 Introduction

Sustainability has become a major concern of the European industrial landscape, driven by consumer demands, regulation and reporting standards, and efficiency and cost. Most emissions are caused by operations in the supply chain [1], which are among the biggest levers for the European industries to become sustainable in the long run. To obtain the right level of sustainability, a company needs to be able to assess how sustainable it is. But existing research underlines the lack of formal models and standardisation especially around product life cycle information [2]. This hinders the measurement of sustainability, despite the available

metrics and measurement tools. The issue is exacerbated by the frequent presence of inadequate information infrastructure, which fails to support every stage of the product life cycle. As a result, information exchange at the interfaces between domains (i.e. between product design, engineering and manufacturing functions) and across company borders (i.e. concerning all supply chain partners in a network) fails to overcome data silos, and the information to properly assess sustainability is often missing or incomplete [3].

Industry 4.0/5.0 technologies are enablers of sustainable manufacturing, particularly addressing information transparency in the context of supply chains. The Digital Product Passport (DPP) [4] emerges as a possible solution to guarantee that the correct information about products and materials can be specified and maintained. DPP is promoted and regulated by the European Commission to boost transparency and encourage circularity by sharing product details throughout its lifecycle. Beginning in 2026, the initial lineup of products, starting with renewable batteries, will be required to meet DPP information criteria set out by supporting regulations. In short, a DPP consists in a digital representation of the necessary data about a particular product. Product lifecycle and circularity assessments as well as the operationalisation of circular business models may be supported by their implementation [5].

This research targets the question on how to develop the digital capabilities required for DPPs in the general domain of manufacturing supply chain. In this context, the study explores the opportunities and implications from the perspective of a renowned European manufacturer, while focusing on generalizing our insights to other manufacturers. In particular, we investigate how the scientific aspects of the different product lifecycle phases can be supported by static and dynamic data, and different levels of granularity: from raw materials to batches and components, to the final product. We made a thorough analysis of the common existing processes and information systems to understand the processes that may be affected by the DPPs, and the typical information systems that can serve as information sources for them at the manufacturer. Following this architectural analysis, we perform a gap analysis, considering the current state-of-affairs at the manufacturer and the DPP information requirements. As a result, we propose a reference architecture for developing DPP systems, and we develop a proof-of-concept of such architecture to support the validation through a panel of experts on real use cases.

This paper is structured as follows. Section 2 motivates the research on Sustainability in Manufacturing Supply Chains and explains how sustainability may be supported by Digital Product Passports. Section 3 describes the performed analysis and introduces our Component-based Product Passport approach. Section 4 presents the validation of the approach through a proof-of-concept and expert opinion. Section 5 concludes this paper.

2 Towards Sustainable Manufacturing Supply Chains

Supply Chain Management (SCM) can be defined as the planning and management of all activities involved in sourcing and procurement, conversion, and all

logistics management activities with the purpose of synchronising supply and demand through information and coordination mechanisms [6]. Supply chains have significant environmental impact, with 8 key sectors responsible for over 50% of annual greenhouse gas emissions worldwide [1]. The Greenhouse Gas (GHG) Protocol acknowledges challenges in accurately assessing CO₂ emissions across the entire value chain and recommends using varied calculation methods based on data availability. The overall suggestion is to tailor calculation methods based on data quality, emphasizing that sustainability achievements are directly tied to information quality [3].

Green material purchasing (alongside product design and manufacturing practices) is crucial for sustainable manufacturing supply chains. This involves raw materials, parts, and all supplies contributing to the final product. Life Cycle Assessment may be used to assess the environmental and social footprint of products and associated manufacturing processes. To perform Life Cycle Assessment, it is necessary to collect data on the environmental effects of a product or service during its entire life cycle, from the extraction of raw materials to its disposal. This includes information about the use of resources and raw materials extraction, energy consumption, CO₂ emissions, and waste production as well as product, component and material composition. Focusing on maintaining product information, DPPs may provide an effective solution supporting Life Cycle Assessment (Fig. 1).

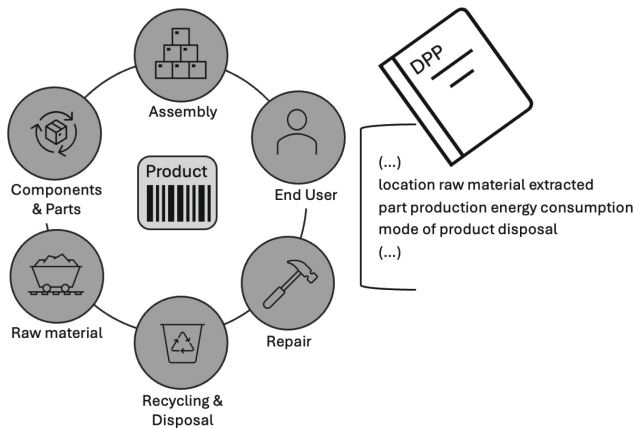


Fig. 1. Digital Product Passports for Supply Chain Lifecycle Assessment

The main idea of a DPP is to enable tracking all associated information of a product, from the conception to the end of life. In other words, DPPs enable transparency and traceability of products and materials through the product value chain, by creating the digital representation and necessary data connections. Thus, product lifecycle and circularity assessments, as well as the operationalisation of circular business models may be supported by DPP implementation [5].

In general, DPPs are applied in 4 product lifecycle phases: product design, manufacturing, operation/maintenance, and recycling. In manufacturing, Product Lifecycle Management (PLM) plays a relevant role for DPP on sustainability data [7], relying on data that are created and modified through the whole product lifecycle.

Information requirements for the DPP can be defined through comparing related research on the topic [8–11], and using industry feedback gathered from the project CIRPASS in consideration of related regulatory initiatives. CIRPASS plays a pivotal role in laying the groundwork for DPPs aligned with relevant standards such as the Ecodesign for Sustainable Product Regulation (ESPR) [12]. The ESPR offers guidance on emissions calculation and gives a first indication of what might be expected from a DPP dataset [9]. The level of granularity for the DPP is specified by the ESPR to be either on product model (e.g. product model A), batch (e.g. product A, from plant X in year 2023) or item (e.g. product A, Serial number 12345678910). Research on product information classification emphasizes that a significant portion of product data is static [11].

The ESPR defines high-level DPP information requirements [9]: Durability, Reliability, Reusability, Upgradability, Repairability, Maintenance, Hazardous Substance Identification, Efficiency, Recycled Content, and Environmental Impact, including recycling and waste generation. Comparing research and the results of CIRPASS, these information requirements extend into detailed categories such as Product Identification, Manufacturer Information, Specifications, Material Composition, Design, Usage, Repair History, and Environmental Certifications [12,13]. These high-level requirements may be refined to arrive at requirements that are more specific to the manufacturer and to the analysis context.

3 Component-Based Digital Product Passport

This section covers the analysis of the current (As-Is) situation of an European manufacturer, presents a gap analysis to note what is currently missing, and introduces a reference (target) architecture that addresses the DPP requirements by proposing data collection and integration capabilities that enable manufacturers to harness the product information from their legacy systems.

3.1 Architectural Analysis of Manufacturing Supply Chain

The manufacturer in our research is an emblematic European-based enterprise that manufactures construction products through discrete manufacturing and assembly, by employing a high-mix and low-volume production strategy. This choice allowed us to understand the challenges associated with producing a diverse product range in smaller quantities, which poses unique considerations and demands, especially in the context of the DPP use case when compared to high-volume production with limited product variations. We focused on two key processes at the manufacturing plant: a) material flow and b) change management.

Material Flow covers the different steps material goes through until it becomes part of the final product. Deliveries can comprise raw materials or components that need to go through one or more manufacturing steps at the plant first, for instance heat treatment or different steel cutting procedures, before being assembled into the final product. Moreover, deliveries can also comprise of components and spare parts that directly go into the product, for instance screws and electronic components (such as batteries). Most components and spare parts delivered to the plant pass through incoming quality control.

Components move through the plant into smaller batches, starting from batches on palettes to smaller handling units. Subsequently, materials are either batch-processed through different machining stages and then repacked into smaller handling units for transportation to assembly lines; or materials move directly into warehousing and repacking. Once at the assembly line, components are assembled into the final product, which is then stored in the warehouse before being shipped to distribution centres.

Most of the processes with regards to material flow occur within the warehouse, while the remaining steps are the responsibility of in-house logistics, production, or assembly. There are three key supporting systems related to the flow of materials: Enterprise Warehouse Management (EWM), Enterprise Resource Planning (ERP), and Manufacturing Execution System (MES). Data is mostly centralised in the ERP, with information related to both inbound quality control and production orders. The EWM manages warehouse and in-house logistics processes such as supply to production and assembly lines, tracking batch locations and transactions. The MES supports production and assembly processes, sharing information with the ERP.

In scenarios where a batch of components is identified as faulty (this can happen at different stages of the material flow, including faults registered by a supplier and internal production issues), it becomes essential to trace their final destination. Therefore, a capability is required to establish ***connections between batches*** from suppliers or internal production, individual components, and the final product. If a batch of components is responsible for issues in final products, it is crucial to not only identify the problematic batch but also the specific set of products in which these components were used. Establishing links between products, components and batches allows for the seamless integration of information about raw materials from suppliers to the final product, ensuring that product information is not solely reliant on static master data such as the bill of materials. Batch level granularity can be achieved mainly through data integration of existing systems. However, achieving item-level granularity for product KPIs entails a significant investment in capabilities, both internally and within the supply chain, including advanced track and trace systems and the placement of further IoT components on shop floor level. Moreover, for manufacturing companies, like the one under study, with relatively low production volumes per product model, the return on investment may not justify the effort required.

Change management involves making alterations to a product model, potentially leading to a new version or revision level. This is crucial for the DPP as it shapes the product model and the foundational information of the actual product. Key stakeholders include internal supply chain participants, like engineering, quality, procurement, health, safety and environmental departments. The main processes are initiated by plant quality and procurement, driven for example by quality issues or supply shortages that can influence product design. This process highlights how new product versions emerge and showcases the complexity of the Product Lifecycle Management (PLM), given the number of stakeholders involved. The manufacturer's high-mix, low-volume strategy results in relatively specific average data per product model due to its low volume. Additionally, the manufacturer's product models have extended life cycles, making changes infrequent.

The ERP, the Computer Aided Design (CAD) system and project management software support this process. Depending on the change required, the CAD system is used to create new drawing of a product of component, while the ERP system is used to register the change in revision. For procurement, the manufacturer uses a cloud-based procurement platform, which allows the connection and collaboration with suppliers and enables streamlining of procurement processes. It is important to note that the usage of the procurement platform varies among the manufacturer's suppliers at present, based on their size and digital capabilities. Specifically for sustainability reporting, a dedicated questionnaire is sent to suppliers to gather additionally necessary information.

3.2 DPP Information Model and Requirements

We propose the high-level information model illustrated in Sect. 3.2 with the respective source systems to address the DPP information requirements in line with typical information structures at manufacturing plants today.

Figure 2 shows the three core systems MES, ERP and EWM, which are the single source of truth for the information, with: a) MES holding information coming directly from production and assembly lines (such as the task completed); b) ERP holding information about the production orders, the orders and suppliers as well as information about the final product; and c) the EWM system is the source for warehouse information, including information about the handling units.

Going in more detail in Fig. 2, the *production order* is a central entity, linking most information of the product. The *serial number* (see within ERP/Product) is the unique ID for a manufactured product, and links information about the *history of the serial number*, which includes any *movements* or status changes registered in the respective IT system. The *Warehouse task* info (see within EWM) links to both the *final product* and the *production order*. Warehouse tasks can be internal movements of products or components, but can also represent the movement of final products to the outside of the plant. *Warehouse task* also relates to an information element called *handling unit*, which represents smaller holding units to transport components and products within the plant,

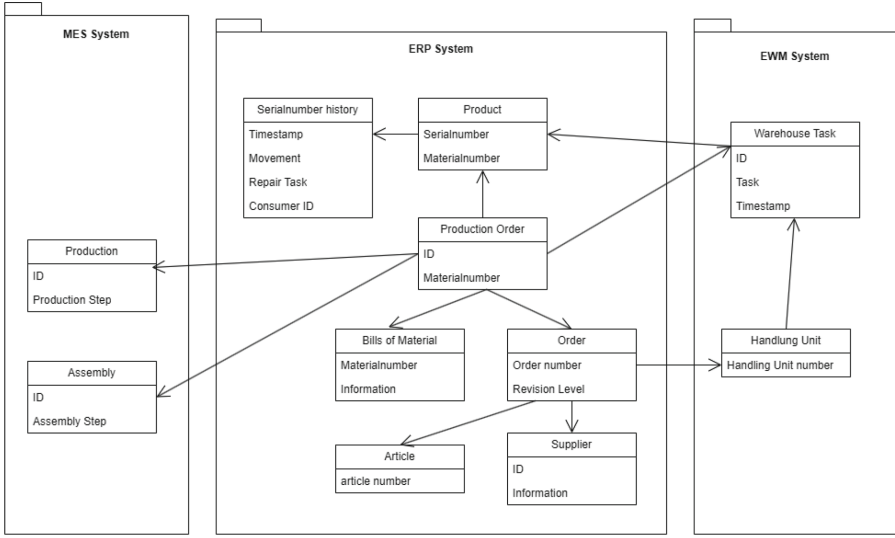


Fig. 2. DPP information model and associated data sources

used by supply production and assembly lines. The *handling unit* is also linked to the *order number* for raw material and components delivered by *suppliers* to the plant. The *order number* in turn relates to information about the *order*, including *supplier* information available in the respective systems. Finally, the *production order* links to information from *production* and *assembly lines* (see within MES for the last two entities). The *material number* is linked to the bill of materials within the respective systems, which lists all materials and parts associated with one product model.

The DPP information model of Fig. 2 adopts a batch-level granularity for data points, such as raw material origin. This makes it possible to consider different product model granularity, besides providing information about the origin of components and materials, which may change periodically based on supplier choices and sourcing. Considering potential future reporting obligations, such as dedicated supply chain regulations, there is a likelihood that this information will require a finer level of granularity. In other words, the batch-level granularity proposed, ensures that individual products can be linked to specific information related to the orders associated with their respective components and materials.

The high level information requirements covered in Sect. 2 were refined into more detailed requirements. Figure 3 shows these information requirements mapped to the phases of the product lifecycle, and also classifying data into static against dynamic data, and direct against aggregated data. Three different combinations are identified: a) Static data collected directly from the source, with the source either the manufacturer's internal system or a system at one of the other participants in the value chain; b) Static data which is aggregated from multiple sources, such as the product environmental footprint, which is partly

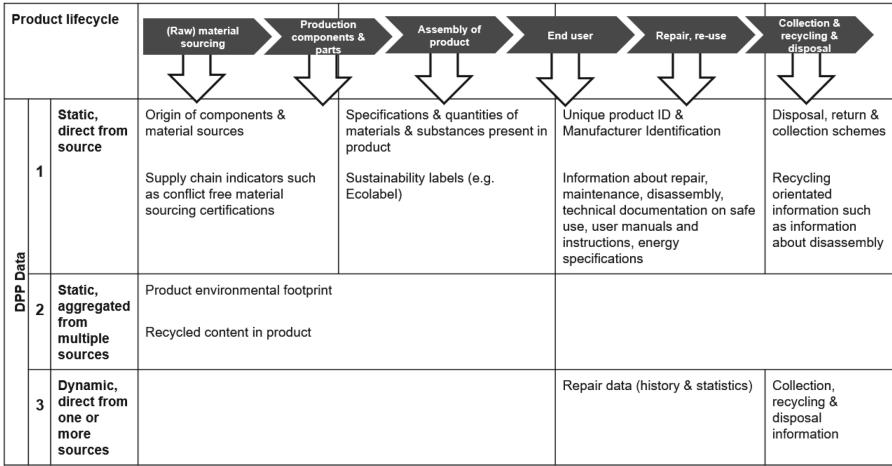


Fig. 3. DPP information requirements aligned to product lifecycle

based on emissions generated in the supply chain prior to the manufacturer in the value chain, and partly based on emissions generated during the manufacturer's internal production and assembly processes; and c) Dynamic data, coming from one or more sources, such as repair data during the use of the product. Service centres external to the OEM may also be allowed to perform certain types of repair services, which means that third parties may also generate dynamic event data about the product in the future.

Static and dynamic data can be distinguished by how regularly they change. Static data usually does not change or changes only seldom, while dynamic data is expected to change at a relevant frequency [14]. Data related to the product's initial lifecycle stages remains relatively stable, and therefore, is static. Data pertaining to usage, repairs, and processes involving product collection, recycling, and disposal are expected to exhibit dynamic behaviour. For static data, a granularity level at the product model or batch level should be adequate due to the limited expected changes over time, resulting in minimal variations among batches. However, adopting a finer granularity at the product item level is possible for dynamic data because movements in and out of repair centers are already tracked based on serial numbers by the manufacturer.

The chosen granularity level imposes specific IT infrastructure requirements: (a) For static data originating from a single source within the manufacturer, suitable interfaces are needed to feed this data into the DPP back-end system; (b) Static data that is aggregated from multiple sources requires a data connection and a data processing layer within the DPP back-end to perform the necessary processing and aggregations; (c) Dynamic data in the later stages of the product lifecycle demands bidirectional data connectivity to allow stakeholders to both read and write data to the DPP system.

We finalize this subsection by presenting an Enterprise Architecture Model¹, resulting from our gap analysis.

3.3 Gap Analysis for DPP System Development

Based on the DPP information model (see Sect. 3.2), the most significant issue regards establishing an internal data collection infrastructure to support DPP information collection. However, the As-Is analysis shows that the existing digital capabilities within the manufacturer present challenges in implementing this infrastructure. Within this context, we identified 3 main gaps to be considered (Fig. 4).

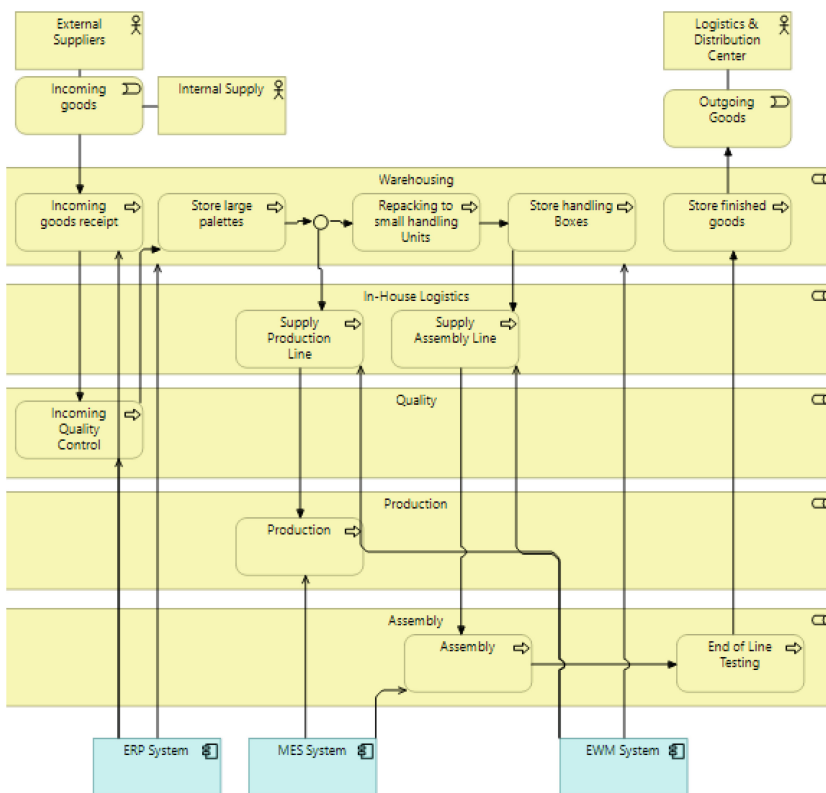


Fig. 4. As-Is: Flow of Materials process and supporting IT systems

¹ We use ArchiMate 3.0.1 (<https://pubs.opengroup.org/architecture/archimate301-doc/>) to model our As-Is EA and our proposed Reference Architecture.

Gap 1: Different Data Granularity Across the Product Lifecycle. The DPP system requires information to be available at product level, whether it is aggregated from batch, model, or item level of that product. Typically, data from supply chain operations is provided to the manufacturer by suppliers in aggregated form, such as X tons of steel delivered with Y kilos of CO₂ emissions generated in the process. In contrast, the manufacturer possesses data at a more granular level, primarily at the production batch level. Therefore, required information for a DPP is also available at the batch level. However, once the product is fully assembled and in use, data is generated at the individual product item level, including usage and repair information on the individual serial number. Information from operations earlier in the product's lifecycle, including the supply chain, production, and assembly processes must be aggregated on product-level. For instance, if a supplier provides data on the amount of steel used, the manufacturer needs to aggregate how much of that steel is incorporated into the final product.

Gap 2: Limited Batch Tracking Capability. The ability to track and trace components individually or in batches within a manufacturing plant is not universally necessary and varies depending on the product being manufactured. In safety-critical domains, products are subject to stringent safety requirements, requiring detailed information on the product's origins and production processes. In such cases, manufacturers are likely to have already implemented internal batch or item tracking systems and can leverage the data for DPPs in more detail. However, for less critical products, comprehensive tracking of components may not be cost-effective or feasible due to the complexity involved, which requires advanced digital competencies, e.g., hardware and software components to support in-house logistics processes. Usually, the mapping of batches to final products relies on manual efforts within the plant, where personnel use ERP and EWM data and timestamps to piece together the sequence of events related to production and assembly. Comprehensive batch tracking through IoT devices like RFID is usually not implemented at the manufacturing plant. Instead, batches and deliveries are matched to final products through the reverse engineering of the chain of events. Either the manual data mapping process needs automation or comprehensive batch tracking should be implemented using IoT devices such as RFID technology at the manufacturing plant to meet DPP information requirements.

Gap 3: Missing Product Data. Missing data can result in several requirements depending on the data source. If the data comes from external partners in the value chain, suitable interfaces for data exchange need to be implemented. Opportunities such as Data Spaces and Blockchain could help with supplier collaboration in the mid to long term future. If the data input is not done properly inside the manufacturer, this likely results in requirements for process optimisation at the time of data creation.

3.4 DPP Reference Architecture

In this section, we propose a reference (target) architecture for the DPP system in manufacturing supply chain, aligning it with existing manufacturing infrastructure and the required digital capabilities. The architecture is illustrated in Fig. 5 through an ArchiMate viewpoint, and is based on [15], which defines the scope and expected outcomes within the broader product lifecycle context. The lifecycle phases are represented as business functions related to the data acquisition capability, and are served by the discussed applications, which interact with data integration services for data exchange. This architecture aims to address the information requirements discussed in Sect. 3.2, and the gaps 1 and 2 presented in the last section. Also, flexibility in operationalising the DPP system is considered, so that third-party service providers can be included for additional benefits of supply chain transparency through material traceability. The decision regarding whether to directly adopt a DPP service from established providers is not within the scope of this research, given the still ongoing evolution of DPP requirements. Instead, the focus is on internal digital capabilities necessary for data collection and integration, which remain essential regardless of whether a third-party or manufacturer system is eventually chosen. It primarily focuses on the manufacturer's internal digital capabilities related to data extraction from production and assembly processes and supporting systems.

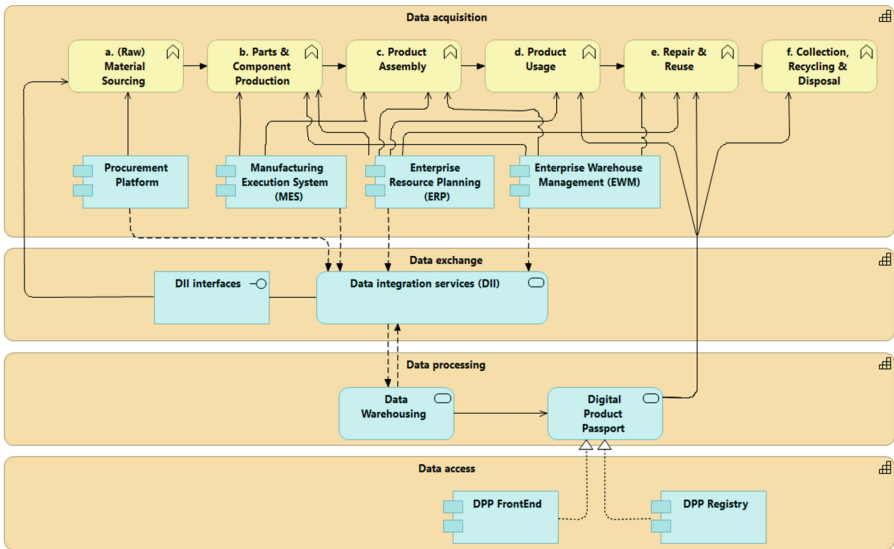


Fig. 5. DPP Reference Architecture for Manufacturing Supply Chain

This architecture addresses the digital capabilities required for the DPP in manufacturing supply chain, giving emphasis to: (1) gathering and mapping

information from the supply chain and (2) integrating data from third parties involved in later stages of the product lifecycle (e.g., external repair centres) into the DPP system. The architecture considers the proposed battery passport [14], and consists of four capability layers: data collection, data exchange, data processing, and data access. The data collection layer gathers information from various sources in the product's lifecycle, especially from early stages suppliers and stakeholders. Data exchange occurs through direct interfaces or the DPP system. The data processing layer encompasses the main DPP back-end, responsible for mapping, processing, and aggregating data from diverse sources. The data access layer includes a DPP repository, necessary for compliance with European Commission policies. The focus is on the Manufacturer's DPP, comprising the back-end system and its connection to the front-end. External interactions with the system can happen through the API or the front-end. Stakeholders in the earlier stages of the product lifecycle provide data through these interfaces, resembling the current scenario where suppliers transmit data to the manufacturer, often through systems procurement platforms. The architecture adopts these design principles:

- **Standardised APIs**, preferably REST-based with JSON-LD payload for interfaces between DPP front-end and back-end. Interfaces between the different back-end components and adjacent systems can be based on current data warehousing and data integration capabilities as well as vendor specific aspects that may influence the choice of service provider for integration components.
- Use of **standardised data models**, preferably based on the Asset Administration Shell (AAS) standard and others relevant for IIoT (e.g., from ETSI, like SAREF) to ensure semantic interoperability and compatibility with services and platforms.
- **Data processing and aggregation** functionalities implemented at data warehouse level with standard interfaces to source systems to ensure compatibility with legacy systems.
- **Microservice architectural style** for data queries and processing, particularly for dynamic data input, to ensure scalability of the system and compatibility with existing IoT pipelines.
- **Event-Driven Architecture** of back-end components to ensure data management and data integrity as well scalability
- **Role-Based Access Control** and authorisation to DPP interfaces.
- Adoption of **existing system for serial number generation**, such as the ERP system, which is usually used for unique identifier generation for data carrier.
- **Automated link between product and batches** either via automated mapping of ERP and EWM] data on data warehouse level, or via life batch tracking system using IoT as next level solution.

At the heart of the proposed architecture lies the standardised representation of products through a standardised data model. The architecture follows the AAS

meta model as baseline for the realisation of the data model [16]. The AAS meta model is considered the standardized digital representation of an asset, aimed at providing interoperability of Industrie 4.0/5.0 components. The architecture recommends data warehousing for legacy system data integration, preferably with structure aligned to standardized models. Service components act as intermediaries connecting the data warehouse, source systems, and the DPP service. Microservices offer flexibility and scalability. Events aid data integrity, serving as notification and data logging for version control of DPP data. Microservices align with DT implementation, ensuring modularity and scalability. The choice of data warehousing for data integration does not contradict a possible decentralised approach to the DPP ecosystem. While data warehousing centralises data storage and management, it does so in a way that complements the broader decentralised architecture. Data warehousing offers a centralised repository for structured data (and can consume data lakes), ensuring data consistency, security, and efficient querying. However, it does not imply centralisation of control or processing. Decentralisation is realized through the microservices and event-driven design. These service components operate independently, handling specific tasks and processing data asynchronously. In addition, the reference architecture fits into the manufacturer's legacy infrastructure and emphasises the importance of interfaces between data sources, the data warehouse, and service components.

4 Validation

4.1 Proof of Concept

The primary goal of the Proof of Concept (PoC) is to showcase the proposed architecture's ability to integrate data. All PoC artefacts described here are available online². The PoC features design core components implemented as Python programs, while the data sources and the data warehouse are implemented through separate SQL-based databases.

The PoC demonstrates how a DPP can be implemented based on manufacturers' legacy infrastructure, i.e., based on the ERP, EWM and MES systems used in the company. Given the proposed granularity levels represented in the reference architecture, the PoC reflects a DPP on product item level (e.g., product with serial number 12345) and subsequent information aggregated on batch level (e.g., batch 1234 delivered by supply X) or individual component level (e.g., component with ID 1234 from supply X), if available. The DPP also requires information to be aggregated on product level based on information on product model level (e.g. material composition, the product contains 1 KG steel from 6 components based on bill of material specifications).

To implement the DPP service as part of the PoC, two python programs are provided. The first establishes the front-end and processes the user input as well as the data that is returned from the second program, which demonstrates the service component responsible for fetching data from the data warehouse

² https://github.com/malina-w/PoC_DPP_public.

component defined in the target architecture. The second program queries the data from the sql database defined by the simplified product representation. This demonstrates the benefits of standardised data representation. Figure 6 shows a screenshot of the PoC frontend, detailing the material breakdown of a particular product.

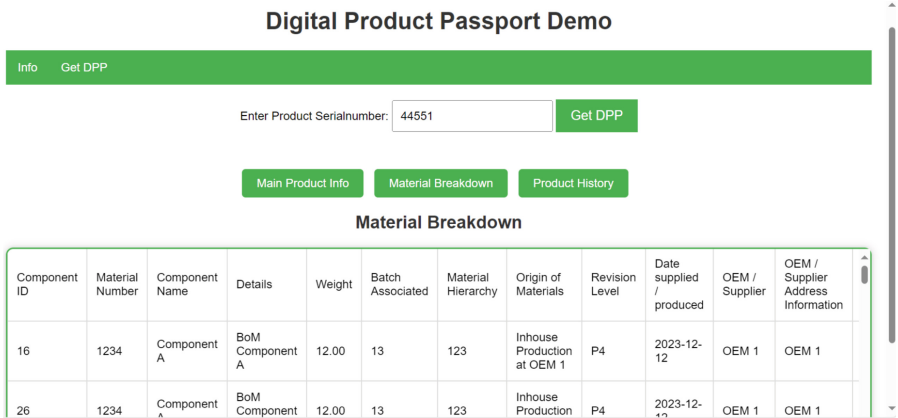


Fig. 6. Screenshot of the DPP system with a product event history

The DPP PoC is compatible with the DPP Information Model and with the Target Architecture. As such, the DPP data model is based on the AAS meta model [16]. To demonstrate the benefits of standardized representation of products, this metamodel is used as basis for the SQL database. The Asset element of the AAS can capture information about the product itself, while submodels define the structure and behavior of the asset. Submodels in the AAS can be utilized to specify the different aspects of the product, such as its physical characteristics, functional capabilities, and lifecycle information.

4.2 Panel of Experts Validation

We conducted a preliminary validation of the proposed architecture and PoC, using a panel of experts composed of 7 people, selected based on their roles and backgrounds at the manufacturer in question, and thus offering diverse perspectives on similar requirements. Example of areas of responsibility of the experts are sustainability, data integration, value creation, quality control, and logistics. The validation comprised of the following 3 steps: a) presentation of the proposed target architecture for the manufacturer’s DPP; b) presentation of the implemented proof of concept; and c) brief interviews and survey with the experts to get their input on how the target architecture and proof of concepts support their requirements and goals. The survey was designed with answers using a typical likert scale with five possible answers, namely: not at all, fairly, moderately, well, and very well.

In general, the survey results show that:

1. The capability of the architecture to integrate, collect and process data was perceived well, with four out of seven experts rating it as ‘very well’.
2. The capability of the proposed IT architecture to merge and aggregate data from different sources (mainly legacy systems) was ‘moderately well scored’, indicating that the demonstrated interfaces need further specifications to operationalise the target architecture for the manufacturers specific systems.
3. The capability of the proposed architecture to support simple and flexible information access scored ‘very well’ by the majority of experts. This indicates that the standardised data model and data warehousing components are perceived as useful by the experts.
4. The capability of the proposed solution to present information with the right level of granularity was rated ‘well’ by five out of seven experts, indicating that the proposed level of granularity for the manufacturer matches the expectations of the experts for the DPP with some limitations, and raising the need for additional attention for data points of the DPP that have not been further implemented into the PoC.
5. The potential of the target architecture to generate future insights (for instance, root cause analysis for quality issues) scored ‘very well’, indicating that stakeholders see the proposed architecture as suitable to generate benefits beyond the DPP use case.

We collected positive opinions and suggestions for improvement from the experts. For example, on the positive side, the machine connectivity expert expressed the chosen microservices design and event-based data management as appropriate because of the compatibility with existing IoT components. Both the data warehouse and interfacing components were pointed out as useful for standardising reporting initiatives across departments, helping to overcome obstacles concerning of individual and manual data flows, often aggregating similar types of information. On the other hand, the expert also commented on the intrinsic complexities of products and components, noting that in the context of DPPs, a product may well be a component from the manufacturers perspective. Thus, the classification of an asset being a product or a component may be relative. Additionally, more detailed information such as time periods and quantity ranges are necessary in addition to information such as serial number range to indicate which products and batches are impacted by a quality incident and should also be considered in the DPP.

5 Conclusion

The Digital Product Passport (DPP) initiative endorsed by the European Commission aims to ensure accurate product information throughout its lifecycle, promoting transparency and circularity, and thus leading to more sustainable supply chains. This paper focuses on how to develop the digital capabilities necessary for DPPs in such context. We investigated how information can be

acquired from the different product lifecycle phases to address the DPP information requirements. We proposed a model mapping each phase to 3 levels about the nature of the DPP data, namely: (1) static, direct from the source; (2) static, aggregated from multiple sources, and (3) dynamic, direct from one or more sources. In addition, we performed an architectural analysis of common processes and information systems in manufacturing supply chains, and we identified the main information sources for the DPP. These results culminated in a proposed reference architecture and a proof-of-concept on a DPP system development for manufacturing supply chain, which was validated through expert panels. This DPP reference architecture is a component-based approach that addresses common gaps like data granularity and batch tracking issues. This approach enables manufacturers to tackle sustainability concerns related to DPP in resource-heavy supply chains on the basis of common legacy infrastructure.

The main limitations and future research were identified as follows. Firstly, following the presented preliminary validation, more concrete results are expected once the product is effectively implemented at the manufacturer. Secondly, the interaction between the DPP and environmental management systems should be investigated more closely. Thirdly, more research is needed on the system's interaction with external parties, leaving opportunities for data ecosystems such as International Data Spaces. In this context, questions emerge about the business case for comprehensive supply chain traceability, particularly in understanding manufacturers' motivations to participate in such initiatives. Finally, further investigation is required on how to (re)use operational ontologies, particularly standardized ones, to address the DPP information requirements.

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